



# Gradients

learn. build. ship. lead.

## ADVANCED AI ENGINEERING COHORT

Complete Curriculum — 8 Weeks · 8 Capstone Projects

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LLMs · RAG · AI Agents · Multi-Agent Systems · Fine-Tuning · Production AI

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## Week 1

# AI Engineering Foundations

*Core concepts, tooling and best practices to kickstart your AI engineering journey.*

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## Introduction to AI Engineering

- What is AI Engineering
- AI Engineering Lifecycle
- AI Application Architecture
- AI Engineering vs ML Engineering
- AI Product Thinking

## LLM Fundamentals

- Evolution of LLMs
- Transformer Architecture Overview
- Self Attention Mechanism
- Tokens & Tokenization
- Embeddings Fundamentals
- Context Windows
- Inference Pipeline
- Sampling Strategies
- Hallucinations Basics

## Python Environment Setup

- Python Installation
- Virtual Environments
- Package Management
- VS Code Setup
- Jupyter Notebooks
- Environment Variables
- API Key Management
- Requirements Files

## AI API Fundamentals

- OpenAI API Setup
- Claude API Setup
- API Authentication
- Making First API Calls
- Chat Completion APIs
- Streaming Responses
- Structured Outputs
- Error Handling
- Rate Limits

## | Prompt Engineering Foundations

- Prompt Structure
- System Prompts
- User Prompts
- Assistant Messages
- Zero-shot Prompting
- Few-shot Prompting
- Role Prompting
- Prompt Templates
- Prompt Iteration

## | Model Parameters Deep Dive

- Max Tokens
- Temperature
- Top-p
- Top-k
- Frequency Penalty
- Presence Penalty
- Stop Sequences
- Response Formatting
- Deterministic vs Creative Outputs
- Token Usage Optimization

## | AI Engineering Best Practices

- Prompt Versioning
- API Cost Optimization
- Retry Mechanisms
- Logging AI Responses
- Output Validation
- Secure API Usage
- Environment Separation
- Debugging LLM Outputs

## Week 2

# RAG & Semantic Search

*Build retrieval-augmented AI systems that ground LLMs in external knowledge using embeddings, vector databases, and advanced retrieval pipelines.*

## RAG Foundations

- What is Retrieval-Augmented Generation (RAG)
- Why RAG Matters
- RAG vs Fine-tuning
- Components of a RAG Pipeline
- Retrieval Pipeline
- Generation Pipeline
- Knowledge Augmentation
- Limitations of Naive RAG
- Advanced RAG Architectures

## Embeddings & Semantic Search

- What are Embeddings
- Semantic Meaning in Vector Space
- Dense vs Sparse Retrieval
- Cosine Similarity
- Dot Product Similarity
- Embedding Models
- OpenAI Embeddings
- BGE Embeddings
- Instructor Embeddings
- Embedding Dimensions
- Embedding Evaluation

## Vector Databases

- Vector Database Fundamentals
- Similarity Search
- Approximate Nearest Neighbor (ANN)
- HNSW Indexing
- Metadata Filtering
- Hybrid Search
- Index Optimization
- Real-time Indexing
- Pinecone
- Qdrant
- Weaviate
- ChromaDB
- FAISS

## | Data Ingestion Pipelines

- PDF Parsing
- Text Extraction
- OCR Pipelines
- Data Cleaning
- Metadata Extraction
- Document Normalization
- Data Deduplication
- Multi-format Document Handling

## | Chunking Strategies

- Fixed-size Chunking
- Recursive Chunking
- Semantic Chunking
- Sliding Window Chunking
- Parent-Child Chunking
- Chunk Overlap Strategies
- Chunk Size Optimization

## | Building RAG Pipelines

- End-to-End RAG Workflow
- LangChain RAG Pipelines
- Retriever Architecture
- Embedding Pipelines
- Context Injection
- Prompt-Augmented Retrieval
- Multi-query Retrieval
- Conversational RAG
- Streaming RAG Responses

## | Retrieval Techniques

- Dense Retrieval
- Sparse Retrieval
- Hybrid Retrieval
- Re-ranking
- Cross Encoders
- Contextual Compression
- Parent Document Retrieval
- Self-query Retrieval
- Query Expansion
- Query Rewriting
- Hypothetical Document Embeddings (HyDE)

## | Advanced RAG

- Corrective RAG
- Adaptive RAG

- Graph RAG
- Agentic RAG
- Self-RAG
- Hierarchical RAG
- Multimodal RAG
- Fusion Retrieval

## | Context Engineering

- Context Window Management
- Token Optimization
- Context Compression
- Dynamic Context Assembly
- Long Context Handling
- Lost in the Middle Problem

## | Evaluation & Observability

- Retrieval Evaluation
- Relevance Scoring
- Hallucination Detection
- Faithfulness Evaluation
- Groundedness

## | Production RAG Systems

- Scalable RAG Architecture
- Latency Optimization
- Caching Strategies
- Cost Optimization
- Async Retrieval Pipelines
- Monitoring & Observability
- Real-time Knowledge Updates
- Secure Retrieval Systems
- Multi-tenant RAG
- Enterprise Knowledge Assistants

## Week 3

# Agentic Architectures

Learn how autonomous AI agents think, reason, use tools, and execute multi-step workflows in real-world AI systems.

## Agent Foundations

- What are AI Agents
- Agentic AI vs Traditional AI
- Components of an Agent
- Agent Lifecycle
- Autonomous Reasoning Systems
- Single-agent vs Multi-agent Systems
- Real-world Agent Use Cases

## The Agent Loop

- Observe → Think → Act → Reflect Loop
- Planning & Reasoning Cycles
- Iterative Decision Making
- Reflection Mechanisms
- Feedback Loops
- Task Decomposition
- Goal-oriented Execution

## Function Calling & Tool Use

- Function Calling Fundamentals
- Tool Calling with OpenAI APIs
- Tool Calling with Claude APIs
- Structured Outputs
- JSON Schema Design
- Tool Selection Strategies
- Dynamic Tool Invocation
- Tool Chaining

## Building Custom Tools

- Web Search Tools
- External API Integrations
- Database Query Tools
- Python Code Execution Tools
- File Handling Tools
- Calculator Tools
- Email & Automation Tools
- Custom Tool Wrappers

## ReAct & Reasoning Patterns

- ReAct Prompting
- Chain-of-Thought Reasoning
- Thought → Action → Observation
- Multi-step Reasoning
- Planning-based Prompting
- Reflection Prompting
- Self-correction Mechanisms

## Agent Memory Systems

- Short-term Memory
- Long-term Memory
- Conversation Memory
- Vector Memory
- Episodic Memory
- Retrieval-based Memory
- Memory Persistence

## Agent Reliability & Safety

- Error Handling
- Timeout Handling
- Fallback Strategies
- Tool Failure Recovery
- Hallucination Reduction
- Output Validation
- Guardrails for Agents
- Secure Tool Execution

## LangChain Agents

- LangChain Agent Fundamentals
- Agent Executors
- Tools in LangChain
- Custom LangChain Tools
- Conversational Agents
- ReAct Agents in LangChain
- Memory Integration
- Multi-tool Workflows

## Search & Web-enabled Agents

- Tavily Search Integration
- Web Search Agents
- Real-time Information Retrieval
- Search + RAG Pipelines
- Multi-source Knowledge Retrieval
- Browser-enabled Agents



## | Advanced Agent Frameworks

- LangGraph Fundamentals
- CrewAI
- AutoGen
- MCP Fundamentals
- Workflow Orchestration
- State-based Agent Systems

## | Production Agent Systems

- Scalable Agent Architectures
- Async Agent Workflows
- Event-driven Agents
- Monitoring & Observability
- Agent Evaluation
- Latency Optimization
- Cost Optimization
- Human-in-the-loop Systems

## Week 4

# Memory, State & Orchestration

*Build stateful AI agents with persistent memory, graph-based orchestration, and production-ready multi-step workflows.*

## Agent Memory Foundations

- What is Agent Memory
- Why Memory Matters in AI Agents
- Stateless vs Stateful Agents
- Memory Architectures
- Memory Retrieval Strategies
- Context Retention Challenges

## Types of Agent Memory

- In-context Memory
- Short-term Memory
- Long-term Memory
- Episodic Memory
- Semantic Memory
- Procedural Memory
- Vector-based Memory
- Retrieval-based Memory

## Persistent Memory Systems

- Persisting Agent State
- Redis for Short-term Memory
- SQLite for Long-term Memory
- Memory Serialization
- Session Management
- Conversation Persistence
- Checkpointing Agent State
- Memory Expiration Strategies

## State Management for Agents

- Stateful Workflows
- Typed State Management
- Shared Agent State
- State Synchronization
- State Transitions
- State Recovery
- Context Propagation
- Multi-session State Handling

## | LangGraph Foundations

- Introduction to LangGraph
- Graph-based Agent Architectures
- Nodes & Edges
- Typed State Graphs
- Graph Execution Flow
- Stateful Graph Pipelines
- Message Passing Between Nodes
- Workflow Composition

## | Workflow Orchestration

- Multi-step Reasoning Workflows
- Conditional Branching
- Dynamic Routing
- Planner-Executor Architectures
- Sequential vs Parallel Execution
- Event-driven Workflows
- Workflow State Tracking
- DAG-based Orchestration

## | Human-in-the-loop Systems

- Human Approval Flows
- Interruptible Agent Workflows
- Feedback-driven State Updates
- Human Escalation Mechanisms
- Pausing & Resuming Workflows
- Interactive Agent Systems
- Review & Correction Loops

## | Long-running Agent Systems

- Handling Long-running Tasks
- Async Workflow Execution
- Background Task Processing
- Task Queues
- Retry & Recovery Mechanisms
- Fault-tolerant Workflows
- Distributed Workflow Execution

## | Multi-Agent Orchestration

- Multi-Agent Coordination
- Shared Memory Across Agents
- Agent Communication Patterns
- Collaborative Reasoning

## | Production-grade Stateful Architectures

- Designing Stateful AI Systems
- Scalable Agent Infrastructure
- Workflow Persistence
- Observability & Monitoring
- State Debugging
- Cost Optimization
- High Availability
- Secure State Management

## | Advanced Orchestration Patterns

- Graph RAG Workflows
- Reflection-based Agents
- Self-correcting Workflows
- Recursive Agent Loops
- Hybrid Deterministic + Agentic Systems
- Memory-augmented RAG
- Event-driven Agent Systems

## Week 5

# Multi-Agent Systems

*Design and orchestrate teams of specialized AI agents that collaborate, delegate tasks, and solve complex workflows autonomously.*

### Multi-Agent Foundations

- What are Multi-Agent Systems
- Single-agent vs Multi-agent Architectures
- Benefits of Multi-Agent Systems
- Agent Collaboration Principles
- Distributed Intelligence
- Autonomous Coordination

### Multi-Agent Architectures

- Supervisor-Worker Architecture
- Hierarchical Agent Systems
- Peer-to-peer Agent Systems
- Planner-Executor Architectures
- Centralized vs Decentralized Agents
- Event-driven Multi-Agent Systems
- Hybrid Agent Architectures

### Role Specialization

- Researcher Agents
- Writer Agents
- Coder Agents
- Critic Agents
- Planner Agents
- Reviewer Agents
- Evaluator Agents
- Tool-specific Agents
- Domain-specialized Agents

### Inter-Agent Communication

- Inter-agent Messaging Protocols
- Structured Task Delegation
- Agent-to-Agent Communication
- Shared Memory Systems
- Message Passing
- Context Sharing
- Communication Routing
- State Synchronization

## Task Orchestration

- Task Decomposition
- Workflow Coordination
- Dynamic Task Assignment
- Sequential Orchestration
- Parallel Agent Execution
- Dependency Management
- Agent Scheduling
- Workflow Monitoring

## CrewAI Fundamentals

- Introduction to CrewAI
- Crews & Team Structures
- Defining Roles
- Defining Tasks
- Agent Goals & Backstories
- Crew Orchestration Strategies
- Sequential vs Hierarchical Processes
- Tool Integration in CrewAI
- Multi-agent Collaboration in CrewAI

## AutoGen Fundamentals

- Introduction to AutoGen
- Conversational Multi-Agent Systems
- Group Chat Architectures
- Multi-Agent Conversations
- Agent Coordination
- Tool-enabled Conversations
- Human Proxy Agents
- Autonomous Conversation Loops

## Consensus & Reliability

- Failure Handling
- Retry Mechanisms
- Consensus Mechanisms
- Conflict Resolution
- Agent Validation
- Self-correction Loops
- Reflection-based Coordination
- Reliability Patterns
- Hallucination Reduction in Teams

## Shared Memory & State

- Shared Context Management
- Shared Vector Memory
- Persistent Team Memory

- Session-based Coordination
- State Tracking Across Agents
- Collaborative Knowledge Bases

## | Advanced Multi-Agent Patterns

- Debate-based Agents
- Reflection Agents
- Swarm Intelligence
- Recursive Agent Systems
- Critic-Reviewer Loops
- Self-improving Agent Systems
- Agent Societies
- Human-in-the-loop Multi-Agent Systems

## | Production Multi-Agent Systems

- Scalable Multi-Agent Architectures
- Distributed Agent Infrastructure
- Monitoring & Observability
- Async Agent Coordination
- Secure Agent Communication
- Access Control Between Agents
- Production Deployment Strategies

## Week 6

# AI Evaluation, Observability & Guardrails

*Measure, monitor, and secure AI systems with evaluation pipelines, tracing, observability, and production-grade guardrails.*

## AI Evaluation Foundations

- Why AI Evals Matter
- Evaluating LLM Applications
- Offline vs Online Evaluation
- Human Evaluation vs Automated Evaluation
- Benchmarking AI Systems
- Continuous Evaluation Pipelines

## Core Evaluation Metrics

- Accuracy Evaluation
- Relevance Evaluation
- Faithfulness Evaluation
- Groundedness
- Hallucination Detection
- Toxicity Detection
- Bias & Fairness Evaluation
- Consistency Evaluation
- Robustness Testing
- Latency & Performance Metrics

## Retrieval & RAG Evaluation

- Retrieval Quality Evaluation
- Precision@K
- Recall@K
- MRR
- NDCG
- Context Relevance
- Answer Relevance
- Retrieval Faithfulness
- RAGAS Framework

## LLM-as-a-Judge Systems

- LLM-as-a-Judge Fundamentals
- Pairwise Evaluation
- Reference-based Evaluation
- Rubric-based Scoring
- Automated Response Grading
- Multi-criteria Evaluation



- Judge Prompt Engineering
- Evaluation Bias & Limitations

## | Evaluation Pipelines

- Automated Evaluation Pipelines
- Batch Evaluation Systems
- Regression Testing for AI
- Evaluation Dataset Design
- Synthetic Test Data Generation
- Continuous Integration for Evals
- Evaluation Reporting Dashboards

## | LangSmith Observability

- Introduction to LangSmith
- Tracing LLM Workflows
- Debugging Agent Pipelines
- Prompt Flow Visualization
- State Inspection
- Execution Monitoring
- Performance Analysis
- Failure Diagnosis

## | Experiment Tracking & Analytics

- Helicone Fundamentals
- Weights & Biases (W&B)
- Experiment Tracking
- Prompt Versioning
- Model Comparison
- Cost Analysis
- Token Usage Tracking
- Latency Monitoring
- AI Analytics Dashboards

## | Guardrails & AI Safety

- What are Guardrails
- Output Validation
- Structured Output Enforcement
- Content Filtering
- Prompt Injection Protection
- Jailbreak Prevention
- Policy Enforcement
- Sensitive Data Protection
- Moderation Pipelines
- Safe Tool Execution

## | Red Teaming AI Systems

- Red Teaming Fundamentals
- Adversarial Prompt Testing
- Edge Case Discovery
- Failure Mode Analysis
- Stress Testing AI Systems
- Safety Benchmarking
- Attack Simulation
- Vulnerability Detection

## | Reliability Engineering for AI

- Production-grade Reliability
- Retry & Fallback Strategies
- Timeout Handling
- Error Recovery
- Deterministic Workflows
- Human-in-the-loop Validation
- Reliability Metrics
- Failure Monitoring

## | Production Observability

- AI Monitoring Systems
- Real-time Alerting
- Distributed Tracing
- Workflow Observability
- Agent Monitoring
- Cost Optimization
- Drift Detection
- Incident Analysis

## Additional

# Fine-Tuning & Model Customization

*Customize and optimize large language models for domain-specific tasks, improved accuracy, and production-grade AI applications.*

## Fine-Tuning Foundations

- What is Fine-tuning
- Fine-tuning vs RAG
- When to Fine-tune
- Full Fine-tuning vs Parameter-efficient Fine-tuning
- Supervised Fine-tuning (SFT)
- Instruction Tuning
- Domain Adaptation
- Task-specific Customization

## LLM Fine-Tuning Fundamentals

- Training vs Inference
- Transformer Training Pipeline
- Forward & Backward Pass
- Loss Functions
- Gradient Descent
- Epochs & Batch Sizes
- Learning Rate Scheduling
- Overfitting & Underfitting

## Dataset Preparation

- Dataset Collection
- Dataset Cleaning
- Dataset Formatting
- Instruction-Response Datasets
- Chat Format Datasets
- Synthetic Data Generation
- Data Augmentation
- Train/Validation/Test Splits
- Dataset Quality Evaluation
- Data Deduplication

## Parameter-efficient Fine-Tuning (PEFT)

- PEFT Fundamentals
- LoRA
- QLoRA
- Adapters
- Prefix Tuning

- Prompt Tuning
- Low-rank Adaptation
- Memory-efficient Training

## | Fine-Tuning Infrastructure

- GPU Requirements
- CUDA Basics
- VRAM Optimization
- Mixed Precision Training
- Quantization-aware Training
- Distributed Fine-tuning
- Training Cost Optimization
- Cloud vs Local Fine-tuning

## | Fine-Tuning Frameworks

- Hugging Face Transformers
- Hugging Face Trainer
- PEFT Library
- TRL (Transformer Reinforcement Learning)
- Axolotl
- Unsloth
- LlamaFactory
- OpenAI Fine-tuning APIs

## | Model Selection

- Open-source LLMs
- Llama Models
- Mistral Models
- Qwen Models
- Gemma Models
- Choosing Base Models
- Model Licensing Considerations
- Small vs Large Models

## | Advanced Fine-Tuning

- Instruction Following Optimization
- Multi-turn Conversation Fine-tuning
- Tool-use Fine-tuning
- Function Calling Fine-tuning
- Multi-task Fine-tuning
- Continual Learning
- Domain-specific Adaptation
- Multimodal Fine-tuning

## | Alignment & Preference Optimization

- RLHF Fundamentals
- Reward Models
- Human Preference Data
- DPO (Direct Preference Optimization)
- Preference Alignment
- Constitutional AI Concepts
- Safety Alignment

## | Fine-Tuning Evaluation

- Benchmark Evaluation
- Instruction Following Evaluation
- Hallucination Evaluation
- Response Quality Assessment
- Toxicity & Safety Evaluation
- Human Evaluation
- LLM-as-a-Judge
- Regression Testing

## | Inference Optimization

- Quantization
- GGUF Models
- 4-bit & 8-bit Quantization
- vLLM
- TensorRT-LLM
- ONNX Runtime
- Speculative Decoding
- KV Cache Optimization

## | Deployment of Fine-Tuned Models

- Model Serving
- API Deployment
- FastAPI Integration
- Ollama Deployment
- Hugging Face Inference Endpoints
- Scalable Inference
- Autoscaling
- Monitoring & Observability

## | Security & Responsible Fine-Tuning

- Dataset Security
- Bias in Fine-tuned Models
- Safety Guardrails
- Prompt Injection Risks
- Model Misalignment
- Responsible AI Practices

## **Production Best Practices**

- Experiment Tracking
- Checkpointing
- Model Versioning
- Reproducibility
- Cost Optimization
- Fine-tuning Pipelines
- CI/CD for Models

## CAPSTONE PROJECTS

# 8 Deployable Projects

*Every project is production-ready and portfolio-worthy. You will build, deploy, and present real systems — not toy demos.*

01

**AI-Powered Document Q&A System**

Build a production RAG app that ingests PDFs, indexes them in a vector database, and returns accurate, source-cited answers using semantic search and an LLM.

02

**Multi-Source Research Agent**

Autonomous agent that takes a research query, searches the web, reads documents, synthesises findings, and produces a structured report using ReAct and tool-calling.

03

**Stateful Customer Support Chatbot**

A persistent support bot with Redis-backed short-term memory, LangGraph-powered escalation flows, and human-in-the-loop approval for sensitive responses.

04

**Multi-Agent Content Pipeline**

CrewAI system where specialized agents (researcher, writer, editor, SEO reviewer) collaborate in a supervised workflow to produce publication-ready content.

05

**Fine-Tuned Domain Expert Model**

Fine-tune an open-source LLM (Llama/Mistral) on a domain-specific QLoRA dataset, evaluate with RAGAS and LLM-as-a-Judge, then deploy via FastAPI + Ollama.

06

**LLM Evaluation & Observability Dashboard**

End-to-end eval pipeline with RAGAS scoring, LangSmith tracing, Helicone cost analytics, and a Streamlit dashboard displaying real-time metrics and failure logs.

07

**Agentic Code Review System**

AI agent that reads pull requests, detects bugs, suggests refactors, runs test coverage checks, and posts structured feedback — all via GitHub tool integrations.

08

**Enterprise Knowledge Assistant**

Production-grade, multi-tenant RAG system with role-based access control, hybrid retrieval, guardrails, real-time indexing, and full observability stack.

## Ready to Build the Future of AI?

*This is not another course. This is your launchpad.*

The Advanced AI Engineering Cohort is a hands-on, project-driven program designed to take you from fundamentals to production-grade AI systems. You will ship real projects, work with industry-level tools, and come out as a job-ready AI Engineer.

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